

Why the abelian theory has no gap, in the language of the target box: its coupling does not run

The founding question of this series, why the commutative theory is gapless while the non-abelian one is expected to be gapped, has a precise form once the strong-coupling gap is a box rather than a point at infinity. The continuous-time expansion of the Kogut–Susskind note is group-blind: it uses only that the electric term is diagonal with a least nonzero Casimir, that the plaquette term is bounded and acts on four links, and Gauss’s law. For compact $U(1)$ it gives a gap $\Delta \geq 4\lambda$ with $\varepsilon = g^2/2$, for g^2 above a threshold of the same size as for $SU(3)$, uniformly in the volume. So **both theories are gapped inside the box**. What differs is whether the flow from weak coupling reaches it. For $SU(3)$ the coupling runs,

$$\frac{dg^2}{d\log(a)} = 2b_0g^4 + O(g^6), \quad b_0 = \frac{11}{16\pi^2} > 0,$$

and the conjecture is that it keeps running until it enters the box. For $U(1)$ the pure gauge theory has $b_0 = 0$: the coupling does not run, the weak-coupling theory is the free photon at every scale, and it never reaches the box; the massless Coulomb phase of Guth (Phys. Rev. D 21 (1980) 2291) and Fröhlich–Spencer (Commun. Math. Phys. 83 (1982) 411; both metadata level) is the rigorous form of this statement, and the gapped set of the lattice $U(1)$ theory is (g_c, ∞) with $g_c > 0$. In the language of the critical-coupling note, T2’ for $SU(3)$ is the statement that the renormalization flow has no fixed point between the ultraviolet and the box, and for $U(1)$ the whole weak-coupling region is a line of fixed points. Constants explicit; nothing promoted.

1. The strong-coupling box does not see the group

Run the expansion of the Kogut–Susskind note for compact $U(1)$ with the Wilson term $(2/g^2) \sum_p (1 - \cos \theta_p)$. The electric term is $(g^2/2) \sum_\ell n_\ell^2$, diagonal in the charge basis, with least nonzero eigenvalue $\varepsilon = g^2/2$ per link. The plaquette operator $w_p = e^{i\theta_p} + e^{-i\theta_p}$ has norm 2 and shifts the charge on the four links of p by ± 1 . Gauss’s law, $\sum_{\ell \ni x} \pm n_\ell = 0$ at every site, makes every excited set a union of closed charged loops, so $|S_k| \geq 4$. The per-step factor of the adjacent-growth count is $u = 128e/(g^2(\varepsilon - \lambda))$ with $N \rightarrow 1$, and the rest is verbatim. The abelian theory is therefore gapped, uniformly in the volume, for g^2 above an explicit threshold of the same order as for $SU(3)$, with a gap approaching the flux-loop energy $4 \cdot \frac{g^2}{2} = 2g^2$ in units $\hbar c/a$. This is the confining phase of compact $U(1)$, and it is the same mechanism as for $SU(3)$: electric flux costs energy per link, and Gauss’s law forces flux to close.

2. What differs is the flow

Let $g(a)$ be the effective coupling of the lattice theory at spacing a , defined by any renormalization scheme that is an exact low-energy reduction. The target box of the target-box note is the region $g^2 \gtrsim 4 \times 10^2$ with few-link corrections of controlled local norm, and the Hamiltonian at any scale that lies in it has a gap.

$SU(3)$. Asymptotic freedom gives, at weak coupling, $g^2(2a) = g^2(a) + 2b_0 \log 2 g^4(a) + O(g^6)$ with $2b_0 \log 2 = 0.0966$, so the coupling grows as the scale coarsens. The mass-gap conjecture is the statement that this growth continues, with corrections staying in the class $\mathcal{C}$, until g^2 enters the box; equivalently, that the flow has no fixed point at intermediate coupling. The intermediate region of the position note §6 is the stretch of the flow in which neither the perturbative form of the running nor the box applies.

$U(1)$. The pure abelian theory has no self-interaction of the gauge field, so $b_0 = 0$ and, at weak coupling, $g(2a) = g(a)$ to all orders: the free photon is a fixed point of the flow at every value of the coupling. The weak-coupling theory therefore never moves toward the box. Guth and Fröhlich–Spencer prove that for g small the lattice theory is in a massless phase with

power-law decay of the Wilson loop and a massless photon, and the strong-coupling expansion proves a gap for g large; the gapped set is (g_c, ∞) with $0 < g_c < \infty$, and the transition at g_c is the zero-temperature phase transition that T2' forbids for $SU(3)$.

3. The three earlier isolations of the non-abelian structure, in this language

The position note §4 lists three places where the non-abelian structure enters: the commutator potential of the zero-momentum sector, the absence of a gauge-invariant operator linear in the electric field, and the one-loop valley potential. All three are statements about the first coefficient of the running: the commutator is what makes $b_0 \neq 0$, the valley potential is its finite-volume image, and the absence of a linear gauge-invariant operator is why the photon channel through which the abelian gap closes does not exist. In the box language they are the reasons the $SU(3)$ flow leaves the ultraviolet at all; the open problem is whether it arrives.

4. What this settles and what it leaves

Settled. The gap is a property of the box, and the box is group-blind: confinement at strong coupling is the same for $U(1)$ and $SU(3)$. The difference between the theories is the flow into the box, and for $U(1)$ the flow is absent. “Commutative fields have no gap” means: their coupling does not run, so the weak-coupling theory is a fixed point outside the box.

Left. For $SU(3)$ the flow leaves the ultraviolet with $b_0 > 0$ and the box is at $g^2 \sim 4 \times 10^2$. Whether the flow arrives is the intermediate-region problem, and every quantity in it is now explicit: the starting slope $2b_0 \log 2 = 0.0966$ per doubling, the class $\mathcal{C}$ that has to be preserved, and the box’s edge and tolerance.

5. Consequence for STATE

The founding question has its answer at the level of the current map: the strong-coupling gap is group-blind, the flow is what distinguishes the groups, and the abelian theory’s coupling does not run. The non-abelian mass gap is the statement that the running, which begins with $b_0 = 11/(16\pi^2)$ for $SU(3)$, does not stop before the box.